Emilia Ma

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Education

University of Toronto Sept 2024 – Expected May 2029
B.Sc. in Computer Science — Dean's List Scholar (2025, 2026) GPA: 3.76/4.0
- Coursework: Data Structures & Algorithms, Software Design, Human Computer Interaction, Systems Programming, Computer Systems & Digital Logic, Theory of Computation, Probability & Statistics

Experience

Ontario Public Service | Backend Dev/Junior Technical Analyst Sept 2026 – Present
- Restored the order orchestrator's Bruno backend test suite from 19 failing scripts to 0 across order creation, driver licence, cross domain, and payment workflows.
- Implemented test-runner logic that queries a Spring Boot Actuator endpoint for configuration resolved from active profiles and runs transaction reversal only for confirmed real downstream calls, skipping mocked and unknown cases to prevent false failures from mock-generated transaction IDs.
- Created fault-injection profiles and JUnit tests across successive stages from refund to financial notification, verifying business error handling, system retries, status-update failures, and recovery.

DataAnnotation | AI Trainer — Software Engineering Oct 2026 – Present
- Design challenging bug fix and feature development tasks to expose coding agent failures; develop reference implementations and automated test suites for implementation-based evaluations.
- Evaluate agent reasoning and generated solutions against task requirements, identifying subtle logic errors, flawed implementation decisions, missed edge cases, and unreliable tests.

Projects

Urbanova | ***Python, TypeScript, React, Mapbox, Express, PostgreSQL*** 2026
Hack the Valley: See how proposed Toronto housing could change neighbourhood views, sunlight, and streetcar commutes.
- Led integration from the Python simulation through Express to React, defining data contracts and implementing schema changes at each layer to keep 3D views synchronized with simulation results.
- Built the 505 Dundas transit simulation, modelling passenger trips from residential areas to workplace destinations, streetcar frequency and headways, variable delays, and crowding.
- Developed a 3D replay driven by recorded simulation events, showing moving streetcars, riders walking to stops, passenger counts, and crowding on the interactive map with data visualizations.

Tempo | ***React, TypeScript, Node.js, Express, MongoDB, Better Auth, Railway*** 2026–Present
Full-stack calendar focused on making scheduling faster and more intuitive through simpler interactions.
- Independently designed, developed, and deployed Tempo end to end, implementing account authentication and user-scoped event storage so users can access their own calendars across sessions and devices.
- Engineered a regex parser that outperforms Google and Apple Calendar across 3+ tested scheduling scenarios, recognizing multi-day and combined date-time ranges (with "now" endpoints) in one field.
- Devised a one-click time entry workflow with instant AM/PM toggling for faster event creation.

GirlMath | ***Python, FastAPI, React Native, TypeScript, MongoDB, DigitalOcean, Gemini API*** 2026
AI powered journaling app supporting girls in STEM by tracking confidence, motivation, and learning trends.
- Led backend dev in a 36-hour hackathon, built FastAPI logic and REST endpoints for a 4 person team.
- Integrated the Gemini API to derive confidence, motivation, and learning trends from journal entries.
- Built frontend API client for backend communication and deployed the FastAPI service to DigitalOcean.

Skills

- **Programming Languages:** Java, Python, TypeScript/JavaScript, SQL, C, HTML/CSS, Assembly
- **Technologies & Tools:** Spring Boot, PostgreSQL, AWS, React/React Native, Node.js, Agents/Codex, Express, MongoDB, FastAPI, Maven, JUnit, Git, Expo, Railway, DigitalOcean, Java Swing
- **Languages:** Proficient in English, French, and Chinese